

Predictive conditional alphas

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Abstract: This paper proposes a two-step procedure to back out conditional alphas. We first estimate daily realized exposures to risk factors using high-frequency data. In the second step, we then retrieve conditional alphas by estimating the conditional expectation of stocks' risk-adjusted returns using daily data. We apply our procedure to a large cross-section of US stock returns and find that conditional alphas carry nontrivial information about future returns. We document this short-term predictability both statistically and economically by sorting stocks on conditional alpha estimates. Finally, we also show that, in contrast, sorting either by raw returns or risk-adjusted returns yields very poor performance.

Keywords: asset pricing, conditional CAPM, realized beta, risk-adjusted performance.

1 Introduction

The unconditional CAPM does not provide a good description of equity markets, failing to explain many market anomalies (see, for instance, size/value/momentum effects in Fama and French, 2012). In fact, the absence of pricing errors does not suffice to ensure a zero unconditional alpha because market betas may change over time in tandem with market volatility and/or with risk premia. Allowing for time-varying betas helps explain most of the (unconditional) value premium given that value stocks are riskiest precisely when risk premium is higher, namely, in recession times (Petkova and Zhang, 2005). In addition, there is a substantial negative bias in the estimation of unconditional alphas due to volatility timing effects (Boguth, Carlson, Fisher and Simutin, 2011).

A conditional CAPM framework typically assumes that alphas and betas are affine on predetermined predictor variables, e.g., stock characteristics, interest rates and spreads as well as other business-cycle indicators. Ferson, Simin and Sarkissian (2008) discuss three stylized factors that emerge from this literature. First, market betas do vary over time in a significant manner (Shanken, 1990; Jagannathan and Wang, 1996; Ferson and Harvey, 1999; Gagliardini, Ossola and Scaillet, 2016), as well as their cross-sectional variation (Cederburg and O'Doherty, 2016). Second, the intercept of the conditional alpha term is smaller than the unconditional alpha. This means that conditional asset pricing models entail on average smaller pricing errors than their unconditional versions. Third, despite of their better fit, conditional asset pricing models still fail in view that conditional alphas are not only nonzero, but also time-varying. See, among others, Wang (2002), Ang and Chen (2007), and Adrian and Franzoni (2009).

We contribute to this literature by taking a different route to estimate conditional alphas and betas. We first employ high-frequency data to retrieve daily realized betas for a given stock (Barndorff-Nielsen and Shephard, 2004; Andersen, Bollerslev, Diebold and Wu, 2006). We then back out the conditional alpha by estimating the conditional expec-

tation of the daily risk-adjusted return (i.e., stock return in excess over the realized risk premia). The resulting alpha estimator is nonparametric, enjoying much more flexibility and robustness than the usual parametric estimators. For instance, it does not require conditional alphas and betas to depend linearly on the conditioning state variables, alleviating considerably misspecification risks. Although we focus on the conditional CAPM in the empirical analysis, the framework is flexible enough to handle asset pricing models with semi-betas (Bollerslev, Patton and Quaedvlieg, 2022), continuous and discontinuous market betas (Todorov and Bollerslev, 2010; Bollerslev, Li and Todorov, 2016), statistical factors (Ait-Sahalia and Xiu, 2017, 2019), and multiple tradable factors (Ait-Sahalia, Kalnina and Xiu, 2020).

Traditionally, the focus is on testing for the absence of systematic pricing errors as in Gibbons, Ross and Shanken (1989). More recently, Grundy and Martin (2001), Lewellen and Nagel (2006), Li and Yang (2011), Ang and Kristensen (2012) and Ma, Lan, Su and Tsai (2020) consider more general settings for testing conditional asset pricing models with time-varying alphas and betas. They advocate for the use of daily data to estimate monthly factor loadings within short local windows. We instead employ ultra-high frequency observations to estimate daily betas. Moreover, we change the focus from *assessing the magnitude of pricing errors* to *exploiting of conditional alphas*. In addition, we also differ from previous nonparametric approaches by assuming that both alphas and betas are measurable functions of conditioning state variables and hence predictable.

We examine the main forces driving conditional alphas in the US stock market from January 2001 to December 2017. We first estimate daily market betas using a multivariate realized kernel with refresh time as in Barndorff-Nielsen, Hansen, Lunde and Shephard (2011), so as to account for market microstructure noise and nonsynchronicity issues. The realized estimator is consistent for daily conditional betas as long as they are constant within the day. Accordingly, we assume a continuous-time process for stock prices in which

the drift and diffusive parameters are measurable functions of conditioning state variables that evolve only at the daily frequency. This ensures that conditional alphas and betas vary over time on a daily basis, but remain constant at any higher (intraday) frequency.

We next compute risk-adjusted returns by deducting from each stock return the risk premia implied by their realized market beta. By definition, the risk-adjusted return is equal to the sum of conditional alpha and idiosyncratic innovation. The key to identify pricing errors is that the conditional expectation of the idiosyncratic term is zero. It thus suffices to estimate the conditional expectation of risk-adjusted returns to obtain conditional alpha estimates. We do this nonparametrically to minimize misspecification and overconditioning risks (Boguth et al., 2011), but we also consider the usual affine specification for the conditional alpha for the sake of comparison.

As predictors for conditional alphas, we employ various interest rates and spreads, market liquidity and volatility measures, and characteristic-based portfolios based on momentum, short- and long-term reversals, size, and value effects. By conditioning on firm-specific attributes, we take the view of Lakonishok, Shleifer and Vishny (1994), Haugen and Baker (1996), and Daniel and Titman (1997) that they could help spot systematic mispricings in equity markets. And, by taking all possible candidates available at the daily frequency as state variables, we also reduce the risk of underconditioning (Ghysels, 1998; Harvey, 2001). Due to their local nature, kernel-based estimators are relatively less prone to persistence issues in the regressors (Ferson et al., 2008). Finally, we also control for data snooping by considering only a few principal components of the various state variables we entertain.

Given conditional alpha estimates, the next step is to assess whether they contain any predictive content given they should convey information about future stock returns if they indeed correlate with pricing errors. We then sort stocks by the conditional alpha estimates to examine whether the stocks in the top quantiles outperform the stocks in the bottom quantiles. The idea is to assess the predictive content of conditional alpha estimates, rather

than implementing a long-short trading strategy. We find that conditional alphas carry nontrivial information on future stock returns. The equal-weighted long-short portfolio based on the top and bottom deciles sorted by affine alphas earn, on average, a strongly statistically positive return of 14.79%, with a standard deviation of 12.21%. Weighting the top and bottom portfolios by value increases the return differential to 15.86%, but also the standard deviation to 15.24%. Sorting by nonparametric alphas yields a slightly worse performance, in that the spread in average returns reduces to 13.00% (or 10.55% if value-weighted), with standard deviation of 12.85% (or 16.06% if value-weighted).

We next assess how much each step in our procedure contributes to this performance relative to sorting by raw returns. The latter entails a long-short portfolio with a very negative return on average, indicating mean reversion in daily returns. Sorting by risk-adjusted returns improve matters, but not much, yielding an average loss of 15.19%. This indicates that eliminating the noise in the risk-adjusted returns by estimating conditional alphas is the crucial step in our sorting strategy. Performance attribution analysis further shows that both portfolios entail significantly positive alphas, despite some exposure to traditional risk factors. Moreover, both performances are quite robust either to excluding stocks with market capitalization in the bottom 5%, or to considering quintiles rather than deciles. In contrast, either increasing the holding period from one day up to 5 days or shifting the signal by one day decreases substantially the performance of both long-short portfolios, especially if sorting by affine alphas. A similar pattern arises if we decrease frequency to weekly or monthly, confirming that the conditional alphas reflect short-lived signals. The long-short strategy based on affine alphas now entails negative average returns, whereas sorting by nonparametric alphas yield small positive average returns.

Apart from the above literature on empirical asset pricing, our paper relates to the statistical literature on the nonparametric (or model-free) estimation of diffusion processes using high-frequency data (see, among others, Bandi and Phillips, 2003; Ait-Sahalia and

Mykland, 2004; Ait-Sahalia, Fan and Xiu, 2010; Comte, Genon-Catalot and Rozenholc, 2012; Bandi and Renò, 2018). In particular, it builds heavily on previous results about how the estimation of quadratic variation functionals affects subsequent statistical inference, as in Corradi, Distaso and Swanson (2009), Corradi, Distaso and Fernandes (2012, 2019) and Bandi and Renò (2018). Finally, it also has a close connection to the statistical literature on factor models in continuous time (see, among others, Mykland and Zhang, 2006; Ait-Sahalia and Xiu, 2017, 2019; Pelger, 2019; Sun and Xu, 2022; Ait-Sahalia, Jacod and Xiu, 2025).

The remainder of this paper proceeds as follows. Section 2 spells out the assumptions we make on the continuous-time multivariate process that governs the dynamics of stock prices. Section 3 establishes the asymptotic properties of the conditional alpha estimator, controlling for the fact that we estimate conditional betas using a realized approach. Section 4 examines the predictive ability of the conditional alphas we estimate for a large cross-section of US equity returns. Section 5 offers some concluding remarks. We collect supplementary material in the appendixes. Appendix A contains all technical proofs, whereas Appendix B entertains a multivariate realized kernel estimator that accounts for both nonsynchronicity and market microstructure noise, as well as the asymptotic behavior of the nonparametric estimator of Treynor and Black’s (1973) appraisal ratio. Finally, we tabulate additional results in Appendix C.

2 Conditional factor pricing model

This section proposes a consistent two-step procedure to identify and estimate conditional alphas. We first obtain daily realized betas for each stock using intraday observations. Realized betas are model-free estimators that, as the time interval shrinks, converge under mild conditions to a well-defined probability limit driven by the covariation process between stock and factors and the quadratic variation process of the factors (Barndorff-Nielsen and

Shephard, 2004). The functional form of this probability limit depends on the continuous-time multivariate process that jointly determines the dynamics of stock prices and factors. We must then ensure that the probability limit of the realized betas is indeed the vector of factor loadings in the discrete-time multifactor model. A necessary condition is that the discrete-time multifactor model results from the exact discretization of the underlying continuous-time multivariate process.

Given realized betas, we adjust daily stock returns by their risk premia. We then carry out a nonparametric regression of the daily risk-adjusted returns on state variables to estimate conditional alphas. Note that realized beta consistency depends on infill asymptotics (i.e., large number of intraday observations), whereas consistency of the conditional alpha estimator rests on long-span asymptotics (i.e., large number of trading days). Given the interplay between infill and long-span asymptotics, we must pay careful attention to the rates at which the number of intraday observations and the number of trading days grow.

2.1 From continuous to discrete time

Let $P_i(s)$ and $\mathbf{F}(s)$ respectively denote the log-prices at time s of asset $i \in \{1, \dots, N\}$ and of the k_F factor-mimicking portfolios that drive assets' excess returns. We assume that both follow continuous-time diffusion processes, with drift and volatility parameters evolving in discrete time as measurable functions of $k_C < \infty$ state variables $\mathbf{C}_t$.

We do not start from the continuous-time version of the CAPM as in Mykland and Zhang (2006), for otherwise the CAPM would not hold in discrete time (Longstaff, 1989). More precisely, for any $s \in [t, t + 1]$ (say, within a trading day t) and $i \in \{1, \dots, N\}$,

$$dP_i(s) = \mu_{i,t} ds + \boldsymbol{\Sigma}'_{i,t} d\mathbf{W}_F(s) + \sigma_i(s) dW_i(s) \quad (1)$$

$$d\sigma_i^2(s) = \kappa_i(\theta_i - \sigma_i^2(s)) ds + \zeta \sigma_i(s) d\widetilde{W}_i(s) \quad (2)$$

$$d\mathbf{F}(s) = \boldsymbol{\mu}_{F,t} ds + \boldsymbol{\Sigma}_{F,t} d\mathbf{W}_F(s), \quad (3)$$

where $\mu_{i,t} \equiv \mu_i(\mathbf{C}_t)$ and $\boldsymbol{\mu}_{F,t} \equiv \boldsymbol{\mu}_F(\mathbf{C}_t)$ are drift functional parameters, $\boldsymbol{\Sigma}_{i,t} \equiv \boldsymbol{\Sigma}_i(\mathbf{C}_t)$ is a

$k_F \times 1$ vector that determines the exposure of asset i to each risk factor, $\Sigma_{F,t} \equiv \Sigma_F(\mathbf{C}_t)$ is the $k_F \times k_F$ covariance matrix of the common risk factors, $\mathbf{W}_F(s)$ is a k_F -dimensional standard Brownian motion, and $W_i(s)$ is a standard Brownian motion independent of $\mathbf{W}_F(s)$. In turn, the idiosyncratic volatility $\sigma_i(s)$ follows a Feller square-root process with standard Brownian motion $\widetilde{W}_i(s)$ independent of $\mathbf{W}_F(s)$ and $\mathbf{C}_t$. We allow for leverage effects by setting $\text{corr}(dW_i(s), d\widetilde{W}_i(s)) = \rho_i \leq 0$.

We next examine whether asset and factor prices are continuous-time semimartingale processes. This is crucial because semimartingale prices are not only consistent with no-arbitrage restrictions, but also necessary in the infill asymptotic theory.

Lemma 1: Let $\mathbf{X}_i(s) = (P_i(s), \mathbf{F}(s))$ evolve as in (1) to (3). Let also $\mathbf{C}(s) = \mathbf{C}_t$ for any $s \in [t, t+1)$ and define the filtration $\mathcal{F}_C(s) = \sigma(\mathbf{C}(\tau), \tau \leq s)$ for $s > 0$. If $\mathbf{C}(s)$ is independent of $W_i(s)$, $\widetilde{W}_i(s)$ and $\mathbf{W}_F(s)$, then $\mathbf{X}_i(s)$ is a conditional semimartingale with independent increments given $\mathcal{F}_C(s)$.

The independence between each Brownian motion and $\mathbf{C}(s)$ ensures that $\mathbb{E}[\Sigma'_{i,t} W_i(s)] = \mathbb{E}[\Sigma_{F,t} \mathbf{W}_F(s)] = 0$. This does not mean that $\mathbf{F}(s)$ and $\mathbf{C}(s)$ are independent, though. The risk factors depend on the state variables for any $s \in [t, t+1)$ through their drift and diffusion parameters. In addition, it follows from Lemma 1 that, given $\mathbf{C}_t$, the continuous-time process $\mathbf{X}_i(s)$ has independent increments for any $s \in [t, t+1)$. As a result, any autocorrelation pattern within the interval $[t, t+1)$ is due exclusively to market microstructure effects. In contrast, due to the dependence on $\mathbf{C}_t$, autocorrelation could arise for daily returns even in the absence of microstructure noise.

Suppose we have M equidistant observations within a day starting at time t , namely, $P_{i,t+\ell/M}$ and $\mathbf{F}_{t+\ell/M}$ with $\ell = 0, \dots, M-1$. We define the vector of realized betas as

$$\widehat{\beta}_{i,t,M} = \left[\sum_{\ell=0}^{M-1} (\mathbf{F}_{t+\frac{\ell+1}{M}} - \mathbf{F}_{t+\frac{\ell}{M}})(\mathbf{F}_{t+\frac{\ell+1}{M}} - \mathbf{F}_{t+\frac{\ell}{M}})' \right]^{-1} \sum_{\ell=0}^{M-1} (\mathbf{F}_{t+\frac{\ell+1}{M}} - \mathbf{F}_{t+\frac{\ell}{M}})(P_{i,t+\frac{\ell+1}{M}} - P_{i,t+\frac{\ell}{M}}). \quad (4)$$

Barndorff-Nielsen and Shephard (2004) show under mild regularity conditions that

$$\text{plim}_{M \rightarrow \infty} \widehat{\beta}_{i,t,M} = \Sigma_{FF,t}^{-1} \Sigma_{F,t} \Sigma_{i,t} \equiv \beta_{i,t}, \quad (5)$$

where $\Sigma_{FF,t} = \Sigma_{F,t} \Sigma'_{F,t}$. As the state variables $\mathbf{C}_t$ are the only drivers of the daily factor loadings, in that $\beta_{i,t} \equiv \beta_i(\mathbf{C}_t)$, betas are constant within a day. This is similar in spirit, even if much milder, to Lewellen and Nagel's (2006) assumption of constant monthly/quarterly betas. In addition, it ensures that realized betas converge to conditional betas without any need for further conditioning.

For simplicity, we henceforth assume without loss of generality that $\Sigma_{FF,t}$ is diagonal. The estimator in (4) assumes that we observe prices and factors without noise. However, there is ample evidence that market microstructure noise is present at the high frequency. As a remedy, one should employ a realized beta estimator that is robust to such a contamination (Ait-Sahalia et al., 2010; Barndorff-Nielsen et al., 2011). This is what we do in the empirical part, using a multivariate realized kernel approach. For notational simplicity, we relegate the case of microstructure noise robust estimators to Appendix B.1.

We now move to discrete time by letting $r_{i,t+1} \equiv \int_t^{t+1} dP_i(s)$ and $\mathbf{f}_{t+1} \equiv \int_t^{t+1} d\mathbf{F}(s)$ respectively denote continuously-compounded stock and factor-mimicking portfolio returns over the time interval $[t, t+1)$. Exact discretization of (1) to (3) yields

$$r_{i,t+1} = \mu_{i,t} + \Sigma'_{i,t} \int_t^{t+1} d\mathbf{W}_F(s) + \int_t^{t+1} \sigma_i(s) dW_i(s) \quad (6)$$

$$\mathbf{f}_{t+1} = \boldsymbol{\mu}_{F,t} + \Sigma_{F,t} \int_t^{t+1} d\mathbf{W}_F(s), \quad t = 1, \dots, T. \quad (7)$$

Note that $\eta_{i,t+1} \equiv \int_t^{t+1} \sigma_i(s) dW_i(s)$ is such that $\mathbb{E}(\eta_{i,t+1} | \mathbf{f}_{t+1}, \mathbf{C}_t) = 0$ in view that the idiosyncratic volatility is independent of the state variables and risk factors. As such,

$$r_{i,t+1} - \eta_{i,t+1} | (\mathbf{f}_{t+1}, \mathbf{C}_t) \sim \mathcal{N}(\mu_{i,t} + (\mathbf{f}_{t+1} - \boldsymbol{\mu}_{F,t})' \Sigma_{FF,t}^{-1} \Sigma_{F,t} \Sigma_{i,t}, \Sigma'_{i,t} \Sigma_{i,t} - \Sigma'_{i,t} \Sigma_{FF,t}^{-1} \Sigma_{i,t}).$$

Using the beta definition in (5) then yields

$$r_{i,t+1} = \alpha_{i,t} + \mathbf{f}'_{t+1} \beta_{i,t} + \epsilon_{i,t+1} + \eta_{i,t+1}, \quad (8)$$

with $\alpha_{i,t} = \mu_{i,t} - \boldsymbol{\mu}'_{F,t} \boldsymbol{\beta}_{i,t}$ and $\epsilon_{i,t+1} | (\mathbf{f}_{t+1}, \mathbf{C}_t) \sim \mathcal{N}(0, \boldsymbol{\Sigma}'_{i,t} \boldsymbol{\Sigma}_{i,t} - \boldsymbol{\Sigma}'_{i,t} \boldsymbol{\Sigma}_{FF,t}^{-1} \boldsymbol{\Sigma}_{i,t})$.

Interestingly, the presence of instantaneous leverage effects does not affect the conditional betas because the idiosyncratic volatility and idiosyncratic Brownian increments (and hence $\eta_{i,t+1}$ and $\epsilon_{i,t+1}$) are independent of the risk factor and state variables. In fact, the exact discretization of (1)–(3) preserves the link between returns and their integrated variance due to the presence of $\eta_{i,t+1}$ in (8). The contemporaneous correlation is indeed equal to $\rho_i \kappa_i / \xi_i \leq 0$ (see Proposition 1 in Bollerslev and Zhou, 2006), whereas the correlation between integrated variance and past returns decreases at a geometric decay as the lag increases (see their Proposition 2).

It is also easy to appreciate why we must assume that the drifts and factor exposures evolve in discrete time. If the state variables were evolving in continuous time, (6) and (7) would then represent just discrete-time approximations. The probability limit of the realized betas would then differ from the true integrated beta given by the integral of the ratio. This assumption is analogous to imposing constant factor loadings over short-time intervals as in Ait-Sahalia and Xiu (2017, 2019) and Pelger (2019). It is worth stressing however that continuous-time variation would only hurt our identification strategy if the discrete-time approximation were such that $\mathbb{E}(\epsilon_{i,t+1} | \mathbf{C}_t, \mathbf{f}_{t+1})$ differs from zero.

This assumption seems at odds with Andersen, Thyrsgaard and Todorov's (2021) and Andersen, Riva, Thyrsgaard and Todorov's (2023) evidence that the cross-sectional dispersion of assets' exposures to observable factors exhibit intraday variation, especially in days of major announcements. To reconcile our assumption with the empirical evidence, one could think along similar lines to Andrei, Cujean and Wilson's (2023) rational-expectations model of informed trading in which investors have disperse information sets. Throughout the day, dispersion decreases as investors process the information content of the trades. This has three implications. First, the beta estimate of each investor converges to the true beta only by the end of the trading day. Second, the cross-section distribution of beta es-

timates across investors becomes less disperse as trades reveal information throughout the day. Third, variation in investors' expectations is typically higher in days of uncertainty, so that intraday variation should be particularly pronounced in days of uncertainty resolution, such as those featuring important news release. See Antunes Neto (2025) for a similar take and corroborating empirical evidence.

2.2 Retrieving conditional alphas from realized betas

Let $Z_{i,t+1} \equiv r_{i,t+1} - \mathbf{f}'_{t+1}\boldsymbol{\beta}_{i,t}$ denote the daily risk-adjusted return at time $t + 1$. It follows from (8) that $Z_{i,t+1} = \alpha_{i,t} + \epsilon_{i,t+1} + \eta_{i,t+1}$. Identification of the conditional alpha stems from the fact that the conditional expectations of $\epsilon_{i,t+1}$ and $\eta_{i,t+1}$ are zero, so that $\alpha_{i,t} \equiv \mathbb{E}(Z_{i,t+1}|\mathbf{C}_t)$.¹ Accordingly, it suffices to estimate the conditional expectation of the risk-adjusted returns to back out conditional alphas.

Unfortunately, this is infeasible because we do not directly observe conditional betas and risk-adjusted returns. To obtain a feasible estimator, we proceed in two steps. First, we adjust asset returns for risk using realized betas:

$$\begin{aligned}\widehat{Z}_{i,t+1,M} &= r_{i,t+1} - \mathbf{f}'_{t+1}\widehat{\boldsymbol{\beta}}_{i,t,M} = r_{i,t+1} - \mathbf{f}'_{t+1}\boldsymbol{\beta}_{i,t} - \mathbf{f}'_{t+1}(\widehat{\boldsymbol{\beta}}_{i,t,M} - \boldsymbol{\beta}_{i,t}) \\ &= \alpha_{i,t} + \epsilon_{i,t+1} + \eta_{i,t+1} - \mathbf{f}'_{t+1}(\widehat{\boldsymbol{\beta}}_{i,t,M} - \boldsymbol{\beta}_{i,t}) = Z_{i,t+1} - \mathbf{f}'_{t+1}(\widehat{\boldsymbol{\beta}}_{i,t,M} - \boldsymbol{\beta}_{i,t}).\end{aligned}$$

Second, we estimate the conditional expectation $\mathbb{E}(\widehat{Z}_{i,t+1,M}|\mathbf{C}_t)$ of the realized risk-adjusted returns using nonparametric kernel methods. In the next section, we establish under mild conditions that the difference between the feasible and infeasible estimators shrinks to zero as the number of intraday observations M increases.

The fact that we employ a nonparametric regression is in stark contrast with the usual practice of imposing an affine structure for the conditional alpha. The motivation is not only to capture nonlinear effects, but also to alleviate any spurious regression issue that may

¹ Suppose we omit a relevant risk factor that correlates with the set of conditioning variables $\mathbf{C}_t$. Conditional alpha estimates would remain measurable functions of $\mathbf{C}_t$, though also reflecting the part of the omitted factor that correlates with $\mathbf{C}_t$.

arise from the use of persistent conditioning variables (Ferson et al., 2008). In particular, the local nature of kernel regressions breaks some of the persistence of mixing regressors for small enough bandwidths.

The price we pay for more robust alpha estimates comes in the guise of the curse of dimensionality. Typically, the dimension of the vector $\mathbf{C}_t$ is relatively high so as to avoid underconditioning bias. However, nonparametric rates of convergence decrease with the dimensionality of the conditioning state vector. We thus condition the realized risk-adjusted returns on the k largest principal components of $\mathbf{C}_t$ (rather than on the full vector).² This allows us to consider a large number of conditioning state variables without incurring into the usual data mining biases that arise in the search for predictor variables (Ferson et al., 2008).

The idea of summarizing the predictive content of many (state) variables into few principal components is exactly what explains the popularity of factor-augmented predictive models (Bai and Ng, 2006; He, 2023). Our setup is rather different, though. First, although we allow k_C to have an arbitrarily large value, we do not let it to grow with T . Second, we entertain principal components just to reduce the dimensionality of $\mathbf{C}_t$, without specifying any factor-augmented model. Let then $\widehat{\mathbf{PC}}_t = (\widehat{PC}_{1,t}, \dots, \widehat{PC}_{k,t})'$ denote the the $k \leq k_C$ principal component estimates of $\mathbf{C}_t$. Our conditional alpha estimator is

$$\widehat{\alpha}_{i,t} = \widehat{m}_{i,T,M}(\widehat{\mathbf{PC}}_t) = \frac{\frac{1}{Th_T^k} \sum_{\ell=1}^{T-1} \widehat{Z}_{i,\ell+1,M} \mathbf{K} \left(\frac{\widehat{\mathbf{PC}}_{\ell} - \widehat{\mathbf{PC}}_t}{h_T} \right)}{\widehat{g}_T(\widehat{\mathbf{PC}}_t)}, \quad (9)$$

where $\widehat{g}_T(\widehat{\mathbf{PC}}_t) = \frac{1}{Th_T^k} \sum_{\ell=1}^{T-1} \mathbf{K} \left(\frac{\widehat{\mathbf{PC}}_{\ell} - \widehat{\mathbf{PC}}_t}{h_T} \right)$ is the kernel estimator of the density g of the vector of principal components. To circumvent the difficulty of estimating $m_i(\mathbf{pc})$ over

² This is a suboptimal dimension-reduction technique given that it is a particular case of the multiple index model (Park, Sriram and Yin, 2010). There are several more efficient methods for dimension reduction in a nonparametric setting: e.g., sliced inverse regression (Li, 1991) and group-Lasso-type criteria (Huang, Horowitz and Wei, 2010). However, their identification requires a nonconstant conditional expectation. While we conjecture alphas vary over time, it is unfair to rule out constant or even zero alphas beforehand.

regions of low density, we consider a trimmed version of $\widehat{m}_{i,T,M}(\mathbf{pc})$ as in Andrews (1995):

$$\text{tr}_T\{\widehat{m}_{i,T,M}(\mathbf{pc})\} = \widehat{m}_{i,T,M}(c) \mathbf{1} \left\{ c \in \widehat{G}_T(\mathbf{pc}) \right\}, \quad (10)$$

where $\widehat{G}_T(\mathbf{pc}) = \{c : \widehat{g}_T(\mathbf{pc}) > d_T\}$. In the next section, we establish the uniform consistency of $\widehat{m}_{i,T,M}(\widehat{\mathbf{PC}}_t)$ as $d_T \rightarrow 0$ at an appropriate rate. In addition, we also derive the asymptotic distribution of $\widehat{m}_{i,T,M}(\mathbf{pc})$ for $\mathbf{pc}$ such that $\widehat{g}_T(\mathbf{pc}) > \varepsilon > 0$ (i.e., for the case in which no trimming is necessary).

Instead of trimming, one could gauge the magnitude of the conditional alphas relative to the noise level by means of Treynor and Black's (1973) appraisal ratio, which gauges the extra units of unsystematic risk investors have to bear for deviating from a passive management strategy. We explore this alternative in Appendix B.2.

3 Asymptotic theory

In this section, we derive the consistency and asymptotic normality of our nonparametric conditional alpha estimator. We first establish the asymptotic equivalence between $\widehat{\alpha}_{i,t+1}$ in (9) and the infeasible estimator based on the unobservable risk-adjusted returns:

$$\widetilde{\alpha}_{i,t+1} = \widetilde{m}_{i,T}(\widehat{\mathbf{PC}}_t) = \frac{\frac{1}{Th_T^k} \sum_{\ell=1}^{T-1} Z_{i,\ell+1} \mathbf{K} \left(\frac{\widehat{\mathbf{PC}}_{\ell} - \widehat{\mathbf{PC}}_t}{h_T} \right)}{\widehat{g}_T(\widehat{\mathbf{PC}}_t)},$$

and then show that using realized risk-adjusted returns has no asymptotic cost. We require the following conditions to hold.

Assumption A For $i = 1, \dots, N$ and some $\delta > 2$,

- (i) Both $\mu_{i,t}$ and the elements of $\boldsymbol{\Sigma}_{i,t}$, $\boldsymbol{\mu}_{F,t}$ and $\boldsymbol{\Sigma}_{F,t}$ in (1) and (3) are $\mathcal{F}_C(t)$ -measurable and δ -dominated. In addition, the idiosyncratic variance $\sigma_i^2(s)$ is independent of the state variables $\mathbf{C}_t$ for $t \leq s < t + 1$.
- (ii) $\mathbb{E} |Z_{i,t}|^\delta < \infty$ and $\{Z_{i,t+1}, \mathbf{C}_t\}_{t=1}^{T-1}$ is strictly stationary and α -mixing with coefficients α_j such that $\sum_{j=1}^{\infty} \alpha_j^{1-2/\delta} < \infty$.

- (iii) The distribution of $\mathbf{C}_t$ with respect to the Lebesgue measure on $\mathbb{R}^{k_C}$ is absolutely continuous, with a bounded density ϕ that is twice continuously differentiable with bounded derivatives.
- (iv) The multivariate kernel function $\mathbf{K}$ is a k -dimensional product kernel with marginal bounded density K , such that $\int_{\mathbb{R}} xK(x) dx = 0$ and $\int_{\mathbb{R}} x^2K(x) dx < \infty$. In addition, the univariate kernel K has an absolutely integrable characteristic function $\Psi(u) = \int_{\mathbb{R}} \exp(iux)K(x) dx$ such that $\int_{\mathbb{R}} |\Psi(u)| du < \infty$.
- (v) Let $m_i(\mathbf{pc}) = \mathbb{E}(Z_{i,t+1} | \mathbf{PC}_t = \mathbf{pc})$, then $m_i(\mathbf{pc}) g(\mathbf{pc})$ is bounded and twice continuously differentiable on $\mathbb{R}^k$ with bounded derivatives.
- (vi) $\sup_{\mathbf{pc} \in \mathbb{R}^k} |m_i(\mathbf{pc})| < \infty$ and $\int_{\mathbb{R}^k} g(\mathbf{pc})^{1-a} d\mathbf{pc} < \infty$ for some $0 < a < 1$.

Assumptions A(ii)–(iii) refer to the state variables $\mathbf{C}_t$, rather than to their first k principal components. Because each principal component is a linear combination of the $k_C < \infty$ elements of $\mathbf{C}_t$, A(ii) ensures that $\mathbf{PC}_t$ is also α -mixing, with the same mixing size as $\mathbf{C}_t$. The marginal density of each principal component is absolutely continuous because they all result from the direct convolution of the k_C absolutely continuous marginal densities of the conditioning variables. As principal components are mutually orthogonal, their joint density is absolutely continuous on $\mathbb{R}^k$. It then readily follows from A(iii) that $\mathbf{PC}_t$ has a absolutely continuous density on $\mathbb{R}^k$. Assumption A(iv) is standard, holding for the multivariate standard normal kernel we use in the empirical application. Assumptions A(v)–(vi) regulate the smoothness and boundedness of the regression function as well as the thickness of the regressors' tails. For example, if $g(\mathbf{pc})$ is multivariate normal, $\int_{\mathbb{R}^k} g(\mathbf{pc})^{1-a} d\mathbf{pc} < \infty$ holds for any a arbitrarily close to unit.

We next establish the asymptotic equivalence between feasible and infeasible estimators.

Proposition 1: Let the conditions in Lemma 1 and Assumption A hold. If $T, M, h_T^{-1} \rightarrow \infty$,

then $\text{tr}_T\{\tilde{m}_{i,T}(\mathbf{pc})\} - \text{tr}_T\{\hat{m}_{i,T,M}(\mathbf{pc})\} = O_p(d_T^{-1}h_T^{-k}M^{-1} + d_T^{-1}h_T^{-k}T^{-1/2}M^{-1/2})$ uniformly in $\mathbf{pc} \in G_T(\mathbf{pc})$.

It is immediate to see that M must grow at a slower rate than T . This is empirically very important. In addition, the conditions on the rate of growth of M are much weaker than in Corradi et al.'s (2009) Theorem 1. This happens because the estimation error in the betas affects only the dependent variable of the nonparametric regression and, as such, it does not enter in the kernel function.

Proposition 2: Let the conditions in Lemma 1 and Assumption A hold. For $0 < Q < \infty$,

$$\begin{aligned} \left[\int_{\mathbb{R}^k} |\text{tr}_T(\hat{m}_{i,T,M}(\mathbf{pc})) - m_i(\mathbf{pc})|^Q g(\mathbf{pc}) d\mathbf{pc} \right]^{\frac{1}{Q}} &= O_p(d_T^{-1}h_T^{-k}T^{-1/2}M^{-1/2} + d_T^{-1}h_T^{-k}M^{-k}) \\ &\quad + O(d_T^{-2}h_T^2) + O_p(d_T^{-2}h_T^{-k}T^{-1/2}) \\ &\quad + O_p(d_T^{a/Q}) + O_p(d_T^{-2}T^{-1/2}). \end{aligned}$$

This establishes the rate at which the Q -norm of $\hat{m}_{i,T,M}(\mathbf{pc}) - m_i(\mathbf{pc})$ shrinks to zero. The first term on the right-hand side captures the estimation error in the realized betas, whereas the second and third terms correspond to the bias and variance of the kernel regression. In turn, the fourth and fifth terms come respectively from trimming and principal component estimation error. Appendix B.1 shows that the asymptotic results in Propositions 1 and 2 remain valid in the presence of market microstructure noise and nonsynchronous time.

To conduct inference, we derive the limiting distribution of $\sqrt{Th^k}(\hat{\alpha}_{i,t,T} - \alpha_{i,t})$ for every $\mathbf{pc}$ such that $g(\mathbf{pc}) > \varepsilon > 0$. For such values of $\mathbf{pc}$, there is no need for trimming and hence we can estimate the conditional alpha by

$$\hat{\alpha}_{i,T,M} = \hat{m}_{i,T,M,t}(\widehat{\mathbf{PC}}_t) = \frac{\frac{1}{Th^k} \sum_{\ell=1}^{T-1} \hat{Z}_{i,\ell+1,M} \mathbf{K}(\widehat{\mathbf{PC}}_\ell - \widehat{\mathbf{PC}}_t)}{\frac{1}{Th^k} \sum_{\ell=1}^{T-1} \mathbf{K}(\widehat{\mathbf{PC}}_\ell - \widehat{\mathbf{PC}}_t)}. \quad (11)$$

Define now

$$\tilde{\alpha}_{i,t,T} = \tilde{m}_{i,t,T}(\widehat{\mathbf{PC}}_t) = \frac{\frac{1}{Th^k} \sum_{\ell=1}^{T-1} Z_{i,\ell+1} \mathbf{K}(\widehat{\mathbf{PC}}_\ell - \widehat{\mathbf{PC}}_t)}{\frac{1}{Th^k} \sum_{\ell=1}^{T-1} \mathbf{K}(\mathbf{PC}_\ell - \mathbf{PC}_t)} \quad (12)$$

where $Z_{i,t} = r_{i,t} - \mathbf{f}'_t \boldsymbol{\beta}_{i,t}$.

Proposition 3: Let the conditions in Lemma 1 and Assumption A hold. For all $\mathbf{pc}$ such that $g(\mathbf{pc}) > \varepsilon > 0$, if $\max \{T^{1/2}h^{-k/2}M^{-1}, h^{-k/2}M^{-1/2}\} \rightarrow 0$, $Th^{k+4} \rightarrow 0$ and $Th^k \rightarrow \infty$ as $T, M, h^{-1} \rightarrow \infty$, then

$$\sqrt{Th^k}(\hat{m}_{i,T}(\mathbf{pc}) - m_i(\mathbf{pc})) \xrightarrow{d} \mathcal{N}\left(0, g(\mathbf{pc})^{-1} \sigma_i^2(\mathbf{pc}) \prod_{\ell=1}^k \int K^2(u_i) du_\ell\right)$$

where $\sigma_i^2(\mathbf{pc}) = \mathbb{E} \left[(Z_{i,t} - m_i(\mathbf{pc}))^2 \mid \mathbf{PC} = \mathbf{pc} \right]$.

This result ensures that we can construct standard errors for $\hat{\alpha}_{i,T,M}$ by replacing $\sigma_i^2(\mathbf{pc})$ and $g(\mathbf{pc})$ with their sample counterparts.

4 Conditional alphas in the US equity market

In this section, we estimate conditional alphas using US equity data. If they relate to pricing errors, they should predict to some extent future stock returns at the short horizon. We proceed in four steps. First, we compute daily factor loadings for every sufficiently liquid stock in the CRSP database. Second, we estimate their conditional alphas by conditioning risk-adjusted returns on a number of state variables. Third, we sort by conditional alpha estimates to form long-short portfolios of stocks belonging to the top and bottom deciles. Finally, we examine their performances to assess the economic relevance of our conditional alpha estimates.

To estimate daily factor loadings, we must first define which risk factors to consider. We employ a single-factor model using the SPDR S&P 500 exchange-traded fund (SPY) as a tradable proxy for the market portfolio. The main advantage is the ease of interpretation

given that the conditional CAPM is a well-established model. In addition, it is very hard to find intraday data for *tradable* factors other than the market portfolio. We prefer tradable factors because they assimilate in a more timely fashion any change in forward-looking expectations.

The sample period runs from January 2001 to December 2017. After burning the first 500 days to estimate the initial conditional alphas, we end up with 2,651 trading days from January 15, 2003 to December 29, 2017. We collect tick-by-tick data for the SPY and for every stock with enough liquidity trading at AMEX, NASDAQ and NYSE (i.e., code shares 10 and 11 at the CRSP database). We consider only stocks for which we have at least 78 intraday observations (i.e., one observation every 5 minutes on average) on a given day to ensure meaningful realized beta estimates as in Amaya, Christoffersen, Jacobs and Vasquez (2015). We also allow only for a maximum number of 2.5% missing observations for each stock, resulting in a data set of 2,044 stocks (on average, 1,370 stocks per day).

To exploit the richness of the tick-by-tick information, we estimate conditional betas using a realized kernel approach with refreshing time. The latter is very convenient because it is robust to market microstructure noise as well as to nonsynchronous trading (see Appendix B.1 for details). We estimate realized betas using either open-to-close or close-to-close intraday returns. For the latter, we include overnight returns in the realized beta estimation using Chiriac and Voev’s (2011) multivariate rescaling procedure. Figure 1 displays the daily behavior of the cross-sectional quantiles of both realized beta estimates. Including overnight squared returns increases the variation in the realized beta quantiles, especially in the right tail. It nevertheless seems clear that the cross-sectional dispersion of realized betas markedly increases by the end of the sample period.

We next compute daily risk-adjusted returns using open-to-close realized betas.³ To

³ The results are very similar regardless of whether we employ open-to-close or close-to-close intraday returns to estimate realized betas. Accordingly, we relegate the results for the close-to-close realized betas to Appendix C.

obtain conditional alphas, we entertain an expanding window to restrict attention to real-time information. We estimate the conditional expectation of the risk-adjusted returns using either an affine specification or a kernel-regression approach using a multivariate standard Gaussian kernel with rule-of-thumb bandwidths. As for the amount of trimming in (10), uniform consistency requires $d_T = O(h_T^\zeta)$ with $0 < \zeta < 1/4$. Note that, because $h_T \rightarrow 0$ as $T \rightarrow \infty$, trimming becomes more aggressive as ζ declines. We set ζ to 0.20 to enforce only a mild trimming, though we also experiment with other values of ζ in Section 4.2.

The daily frequency of the analysis dictates the choice of state variables. In particular, we consider changes in the VIX index and in the Fed Fund rate, credit and term spreads, momentum, size and value factors, short- and long-term reversal factors, and volatility risk premium (namely, difference between the VIX index and realized volatility).⁴ To avoid the curse of dimensionality, we consider only their three first principal components.

Figure 2 displays the daily cross-sectional quantiles of the conditional alpha estimates. In contrast to the pattern we observe for realized market betas, the cross-sectional dispersion of the conditional alpha estimates is much more erratic, exhibiting a great deal of volatility clustering. More importantly perhaps, there is a huge difference between affine and nonparametric estimates of the conditional alphas. In particular, nonparametric alpha estimates are much larger, especially between 2008 and 2010. Breaking them down by market capitalization quantiles shows that the conditional alpha estimates are, on average, much larger in magnitude for small caps than for big caps. Similar, but slightly less pronounced patterns arise if we sort stocks by liquidity measures, such as trading volume and number of intraday observations.

Figure 3 shows that nonparametric alpha estimates exhibit much less persistence than

⁴ We gather the Fama-French, momentum and reversal factors from Kenneth French's webpage at http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html. Interest rates are from the FRED database, whereas the VIX index is from the Chicago Board Options Exchange.

their affine counterparts. This is due to the fact that kernel estimators are local in nature, so that they do not automatically inherit persistence in the regressors. We find a somewhat similar pattern for realized betas. Adding overnight returns decreases the persistence in the realized betas over time, in line with the empirical evidence that overnight squared returns display little predictability (Bollerslev et al., 2016).

We next examine whether the conditional alpha estimates help predict future returns. Preliminary analysis reveals that, even though they are less persistent, nonparametric alphas entail on average larger correlations than their affine counterparts at different horizons. In particular, affine alphas correlate only contemporaneously with stock returns, whereas the average correlations of the nonparametric alphas are around 2% up to 4 days ahead.

4.1 Do conditional alphas help predict future returns?

The conditional alpha estimates should reflect to some extent sparse, short-lived signals about future returns (Chinco, Clark-Joseph and Ye, 2019). We exploit the information in the cross-section of conditional alphas by sorting stocks at the end of each day into decile portfolios, using only real-time information.

Tables 1 and 2 report descriptive statistics of the daily returns on the decile portfolios based respectively on affine and nonparametric estimates of the conditional alphas. Average returns on alpha-sorted portfolios are approximately nondecreasing along deciles, regardless of the estimation method and weighting scheme. In contrast, standard deviations exhibit a distinctive U-shaped pattern for equal-weighted portfolio returns, which flips to hump-shaped once we weigh stocks by value. This is not surprising given that both market capitalization and number of trades display similar inverted U-shaped patterns.

The bottom and top decile portfolios based on affine alphas yield average returns of 9.27% and 24.07% per year, with robust standard deviations of 28.77% and 25.92% under uniform weights. As expected, weighing by market capitalization compresses both average

returns and standard deviations significantly. Average returns on the bottom and top decile portfolios become 1.09% and 16.95% per year, with standard deviations of 12.53% and 18.79%, respectively. This difference in performances justifies a long-short portfolio that buys stocks with the highest alphas and sells those with the lowest alphas. The last column of Table 1 shows that such a portfolio yields average returns of 14.79% and 15.85% for the equal- and value-weighted long-short portfolios, with reward-to-variability ratios of 1.21 and 1.04, respectively.

The average returns on the equal-weighted portfolios based on nonparametric alpha estimates are somewhat lower: 8.58% for the bottom decile portfolio and 21.57% for the top decile portfolio. Weighing by market capitalization has a huge impact, reducing them respectively to 1.38% and 11.93%. The spreads remain large, though. The average return on the equal- and value-weighted portfolios are of 13.00% and 10.55% per year, with reward-to-variability ratios of 1.01 and 0.66, respectively.

Tables 1 and 2 also display some additional characteristics of the decile portfolios. The stocks in the top decile have, on average, a larger market capitalization than the stocks in the bottom decile. Accordingly, it is hard to rationalize their difference in performance by the size factor. The same applies to the market factor in that the top and bottom deciles have similar realized betas. As for liquidity measures, number of trades and volume seem to exhibit opposite patterns at least for the top decile, whose stocks trade on average more frequently, but at lower dollar volume. Altogether, size, market and liquidity factors do not seem to explain the cross-sectional differences in conditional alphas.

Table 3 reveals that our long-short strategy based on conditional alpha estimates improve over time, reflecting the better precision of the conditional alpha estimates due to the expanding window. More specifically, we split the sample period into two equal parts: January 15, 2003 to June 25, 2010 and June 28, 2010 to December 29, 2017. The long-short portfolios become not only less volatile, but also much more profitable in the most recent

period.

We now ask whether sorting stocks either by raw returns or risk-adjusted returns could yield similar results. The evidence in Table 4 is unequivocal. Average returns are monotonically decreasing along deciles if sorting by raw or risk-adjusted returns. As a result, the long-short portfolio based on raw returns loses on average 20.19% per year, whereas sorting by risk-adjusted returns entails a slightly lower average loss of 15.19%. As both standard deviations are around 17%, these negative returns are highly significant, indicating some sort of short-term reversal, as in Roll (1984). Albeit moving from raw returns to risk-adjusted returns does not improve much average returns, sorting by conditional alphas pays off handsomely, yielding a remarkable wedge of almost 30% per year relative to sorting by risk-adjusted returns.

Next, we ask whether it is possible to reproduce their performance using a combination of passive strategies. In particular, we check whether they have significant exposures to the usual factor-mimicking portfolios: excess market returns (MKT), size (SMB), value (HML), profitability (RMW), investment (CMA), momentum (MOM), short-term reversal (STREV), and long-term reversal (LTREV).⁵ Table 5 reports that the long-short equal-weighted portfolios based on conditional alpha estimates display weak exposure to conventional factors, loading specifically on winning large caps.⁶ More importantly however, they significantly outperform the passive strategy based on conventional factors, with significantly positive intercept estimates regardless of the sample period. This means that the traditional factors cannot fully explain the performances of the long-short portfolios based on conditional alpha estimates, which amount to abnormal returns of approximately 16% and 13% per year for affine and nonparametric alphas, respectively.

⁵ We collect the corresponding factor-mimicking portfolio returns from Kenneth French's webpage.

⁶ This indicates that, apart from short-lived signals about future returns, our conditional alpha estimates might also be capturing some form of misspecification in the risk adjustment.

4.2 Further analyses and robustness checks

We start by looking at whether our findings so far depend heavily on the rebalancing periodicity and/or frequency of the conditional alpha estimates. In the interest of space, we provide just a brief discussion, relegating the full set of results to Appendix C.

The long-short portfolios based on affine alphas perform slightly better than those based on nonparametric alphas with daily rebalancing. Their relative performance flips once we entertain longer holding periods (see Table C1). In particular, average returns on the long-short equal-weighted portfolios based on affine alphas reduce monotonically from 6.50% to 0.43% as we increase the holding period from 2 to 5 days, whereas their volatility decreases much less dramatically from 10.74% to 9.78% per year. In turn, average returns decline much less—from 6.81% to 3.83% per year—for the long-short strategy resting on nonparametric alphas. As before, volatility also decreases less markedly, from 11.21% to 10.28% per year.

We next investigate how the informational content of conditional alphas varies with frequency. To do so, we aggregate daily risk-adjusted returns to compute weekly/monthly conditional alpha estimates. The performance of the long-short portfolios based on affine alphas breaks down as we decrease frequency (see Table C2 and C3). Sorting by weekly affine alphas yields negative returns of -0.55%, for instance. Sorting by weekly nonparametric alphas fare slightly better, with average returns of 1.73%. The same happens if we shift the alpha signal by one day, with possible exception to the value-weighted long-short portfolios resting on nonparametric alpha estimates (see Table C4 and C5). Altogether, this suggests that conditional alphas are best geared to capture high-frequency information that evaporates quickly with the holding period and aggregation frequency.

Excluding micro caps, i.e., stocks with the 5% lowest market capitalization, does not change qualitatively the results (see Table C6). The long-short equal-weighted portfolios still have healthy reward-to-variability ratios around 1. Sorting by conditional alphas into

quintiles also yields very similar quantitative results (see Table C7). The difference in average returns between the top and bottom quintile portfolios remains substantial (10.30% for affine alphas and 9.34% for nonparametric alphas), despite similar standard deviations. The reward-to-variability ratios of the resulting long-short portfolios remain close to 1 for both alphas estimates.

Redoing the entire analysis using close-to-close realized betas does not change the results (see Tables C8 to C17). The performances of the long-short portfolios hardly change if we trim more aggressively (namely, $\zeta = 0.01, 0.10$), though they worsen a bit if we employ no trimming. This happens because conditional alpha estimates are sometimes large only because of a very low value of the denominator in (9). This is exactly the sort of situation that trimming avoids by setting alpha to zero if there is too much uncertainty (i.e., very low density of the conditioning variables). Accordingly, trimming improves performance by putting some discipline in the conditional alpha estimation.

5 Conclusion

This paper proposes a two-step estimator of conditional alphas within a multifactor asset pricing model. We first estimate conditional factor loadings to control for risk exposures. We then regress risk-adjusted returns on conditioning state variables to back out conditional alphas. Despite the simplicity of our two-stage estimator, it requires thinking very carefully about the joint dynamics of factors and stock prices. In particular, we start with a multivariate diffusion process in continuous time for which exact discretization yields the conditional multifactor asset pricing model of interest. To accomplish this, we must assume that the drift and diffusive parameters are measurable functions of state variables that change only in discrete time (say, at the daily frequency). Accordingly, factor loadings are also constant within a day and hence we may estimate them using high-frequency data by means of a realized approach. We provide conditions under which the estimation error

in the realized factor loadings is asymptotically negligible.

To assess empirical relevance, we estimate daily conditional alphas for US stocks. To proxy for the state variables, we employ the principal components of size, value, momentum and reversal mimicking portfolio returns, interest-rate spreads and market volatility measures. If our conditional alpha estimates indeed reflect to some extent pricing errors, they should help predict future stock returns. To exploit the information in the cross-section of conditional alphas, we sort stocks at the end of each day into decile portfolios. We find robust evidence that portfolios that take long positions in top-decile stocks and short positions in bottom-decile stocks yield significantly positive (risk-adjusted) returns. This is in stark contrast with the performance of long-short portfolios that sort either by raw returns or risk-adjusted returns, which entail very small returns in magnitude. Importantly, we view our results just as evidence that the conditional alphas embed information about future returns at the very short horizon. We do not promote the corresponding strategy as feasible, given trading costs would kill any abnormal return given the high daily turnover.

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Figure 1

Daily quantiles of realized betas using open-to-close and close-to-close intraday returns
We plot the daily quantiles as well as top and bottom percentiles and deciles of the realized betas. We estimate the latter using either open-to-close or close-to-close intraday returns. The difference is that close-to-close realized betas include overnight returns using Chiriac and Voev’s (2011) rescaling procedure.

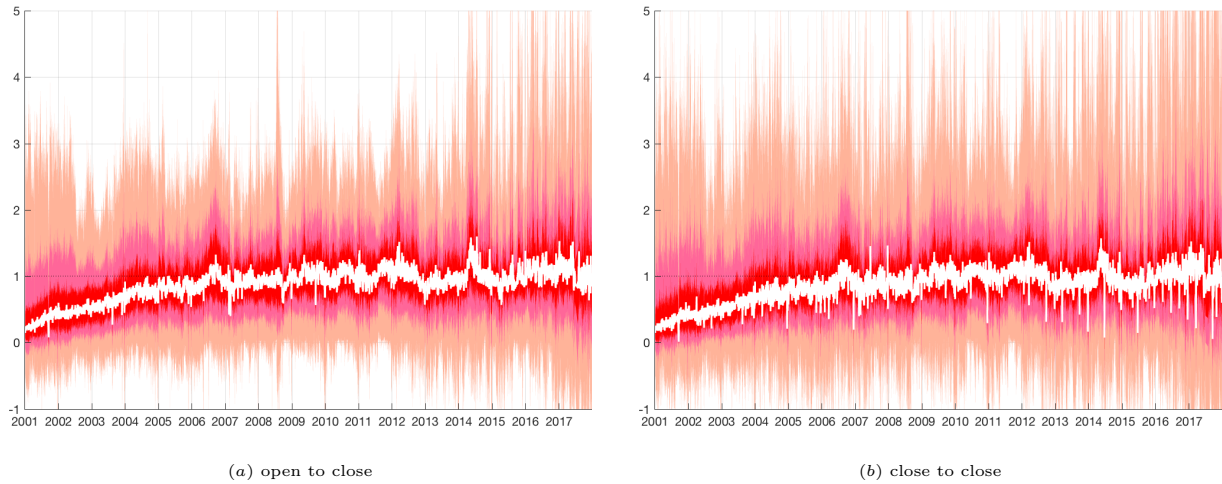


Figure 2

Daily quantiles of conditional alpha estimates

We plot daily quantiles of the conditional alpha estimates using the open-to-close realized betas to adjust returns for risk. Panel (a) plots affine alpha estimates, whereas panel (b) exhibits nonparametric alpha estimates. Panels (c) to (f) break down the latter estimates by market capitalization quantiles.

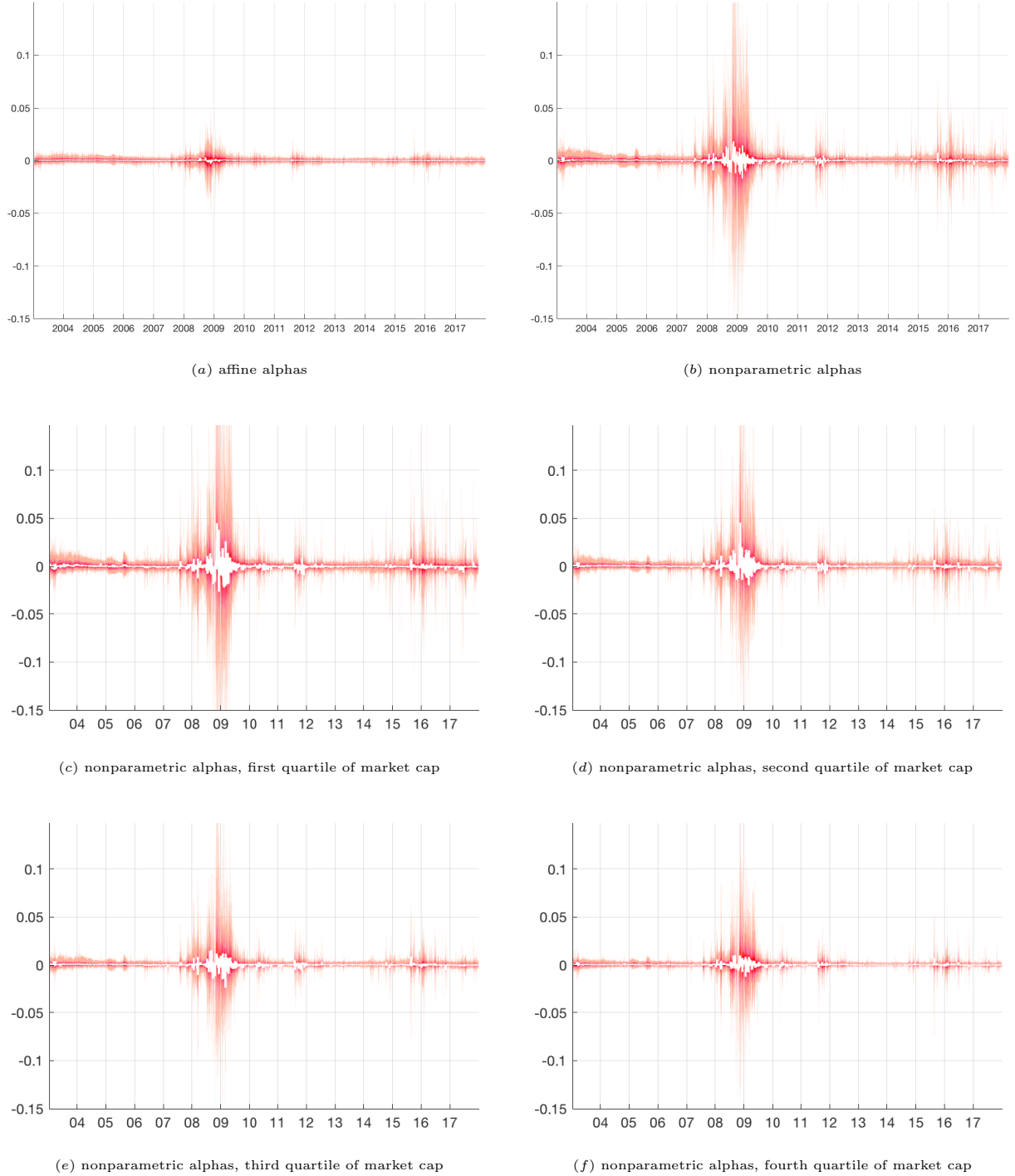
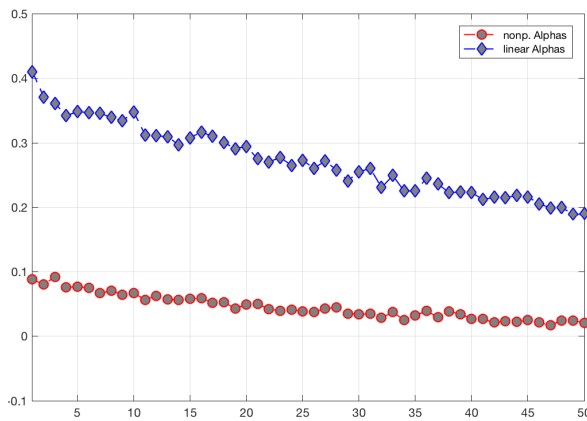


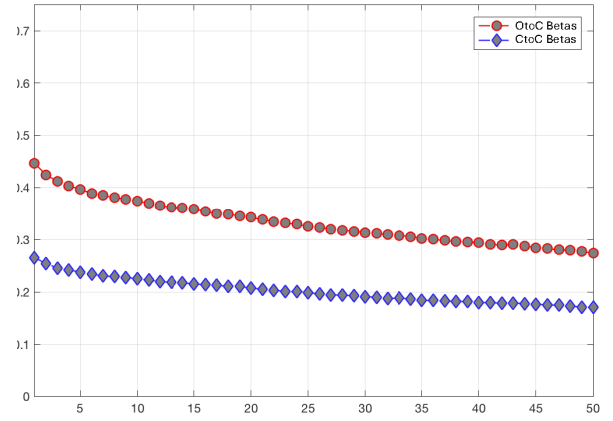
Figure 3

Autocorrelation functions of conditional alpha and beta estimates

We plot the autocorrelation functions of the conditional alpha and beta estimates up to 50 lags. Panel (a) displays the autocorrelation function of the affine and nonparametric estimates of the conditional alphas, whereas panel (b) exhibits the autocorrelation functions of the realized betas using open-to-close and close-to-close intraday returns.



(a) affine and nonparametric alpha estimates



(b) open-to-close and close-to-close realized betas

Table 1 Decile portfolios based on affine conditional alpha estimates

We report annualized mean returns (mret), standard deviations (stddev) and the resulting t -statistics of the decile portfolios based on affine conditional alpha estimates, as well as for a portfolio that buys the stocks in the highest decile and shorts those in the first decile (spread). The sample period runs from January 15, 2003 to December 29, 2017. We also document the average market capitalization (mktcap), number of trades (ntrades) and number of stocks (nstocks) across decile portfolios.

equal-weighted decile portfolios										
	1	2	3	4	5	6	7	8	9	10 spread
mret	9.27	11.81	11.21	13.72	14.31	13.34	15.76	15.71	17.74	24.07 14.79
stddev	28.77	23.97	22.29	21.42	20.97	20.80	21.02	21.62	22.79	25.92 12.21
t-stat	1.24	1.89	1.93	2.46	2.62	2.46	2.88	2.79	2.99	3.57 4.65
mktcap	6.10	10.35	11.30	12.19	12.05	11.90	11.52	10.54	9.07	7.59
ntrades	2,111	2,435	2,504	2,537	2,531	2,538	2,533	2,494	2,457	2,403
nstocks	137	137	137	137	137	137	137	137	137	137
value-weighted decile portfolios										
	1	2	3	4	5	6	7	8	9	10 spread
mret	1.09	-0.23	7.89	7.85	13.17	13.55	13.81	15.05	17.03	16.95 15.86
stddev	12.53	18.86	19.51	21.01	21.77	21.76	22.40	22.68	20.86	18.79 15.24
t-stat	0.34	-0.05	1.55	1.44	2.32	2.39	2.37	2.55	3.14	3.46 3.99

Table 2 Decile portfolios based on nonparametric conditional alpha estimates

We report annualized mean returns (mret), standard deviations (stddev) and the resulting t -statistics of the decile portfolios based on nonparametric conditional alpha estimates, as well as for a portfolio that buys the stocks in the highest decile and shorts those in the first decile (spread). The sample period runs from January 15, 2003 to December 29, 2017. We also document the average market capitalization (mktcap), number of trades (ntrades) and number of stocks (nstocks) across decile portfolios.

equal-weighted decile portfolios											
	1	2	3	4	5	6	7	8	9	10	spread
mret	8.58	12.09	12.51	13.04	13.81	14.62	15.34	17.26	17.96	21.57	13.00
stddev	28.71	23.99	22.26	21.32	20.76	20.71	20.81	21.56	23.11	26.67	12.85
t-stat	1.15	1.94	2.16	2.35	2.56	2.71	2.83	3.07	2.99	3.11	3.88
mktcap	4.91	11.11	12.84	12.86	12.12	11.23	10.99	10.09	8.62	7.83	
ntrades	2,025	2,527	2,595	2,563	2,519	2,456	2,458	2,464	2,480	2,456	
nstocks	137	137	137	137	137	137	137	137	137	137	
value-weighted decile portfolios											
	1	2	3	4	5	6	7	8	9	10	spread
mret	1.38	5.10	9.31	10.42	13.22	13.10	14.24	15.06	12.05	11.93	10.55
stddev	15.24	22.89	21.93	22.58	21.88	21.15	20.94	19.41	17.82	17.74	16.06
t-stat	0.35	0.86	1.63	1.77	2.32	2.38	2.61	2.98	2.60	2.58	2.52

Table 3 Descriptive statistics of the long-short portfolios over time

We report annualized average returns, standard deviations and the resulting t -statistics of the long-short equal-weighted portfolios based on conditional alphas for different sample periods. The top panel display the results for affine conditional alpha estimates, whereas the bottom panel document those for nonparametric estimates. The sample period runs from January 15, 2003 to December 29, 2017; with the first subsample ending on June 25, 2010 and the second subsample beginning on June 28, 2010.

	entire sample	first half	second half
affine conditional alpha estimates			
average return	14.79	11.97	17.61
standard deviation	12.21	14.08	9.99
t-statistic	4.65	2.31	4.79
nonparametric conditional alpha estimates			
average return	13.00	10.80	15.20
standard deviation	12.85	14.02	11.56
t-statistic	3.88	2.09	3.57

Table 4 Decile portfolios based on raw and risk-adjusted returns

We report annualized mean returns (mret), standard deviations (stddev) and the resulting t -statistics of the decile equal-weighted portfolios based on raw and risk-adjusted returns based on open-to-close realized betas. The last column exhibits the statistics for the long-short portfolios that buy stocks in the highest decile and short those in the first decile. The sample period runs from January 15, 2003 to December 29, 2017. We also document the average market capitalization (mktcap) and number of trades (ntrades) across decile portfolios.

sorting by raw returns											
	1	2	3	4	5	6	7	8	9	10	spread
mret	26.86	21.34	18.64	17.75	15.46	12.16	10.80	10.07	7.64	6.67	-20.19
stddev	30.28	24.52	22.41	21.38	20.68	20.38	20.51	21.32	22.85	28.08	17.84
t-stat	3.41	3.34	3.19	3.19	2.87	2.29	2.02	1.81	1.28	0.91	-4.34
mktcap	5.52	9.17	10.95	12.06	12.64	12.68	12.28	11.36	9.80	6.17	
ntrades	2,696	2,432	2,361	2,342	2,337	2,341	2,356	2,395	2,489	2,792	
sorting by risk-adjusted returns											
	1	2	3	4	5	6	7	8	9	10	spread
mret	23.75	19.69	17.88	15.89	14.90	12.21	12.36	11.71	10.40	8.56	-15.19
stddev	30.24	24.48	22.21	21.13	20.35	20.03	20.32	21.29	23.04	28.34	16.46
t-stat	3.02	3.09	3.09	2.89	2.81	2.34	2.34	2.11	1.73	1.16	-3.54
mktcap	5.33	8.97	10.92	12.18	12.66	12.84	12.39	11.47	9.77	6.07	
ntrades	2,670	2,432	2,373	2,352	2,335	2,337	2,356	2,408	2,495	2,782	

Table 5 Performance attribution for portfolios based on conditional alpha estimates

We regress returns on the long-short equal-weighted portfolio based either on affine or nonparametric conditional alpha estimates on a constant, the excess returns on the market portfolio (MKT), and the factor-mimicking portfolios relating to size (SMB), value (HML), profitability (RMW), investment (CMA), momentum (MOM), short- and long-term reversals (STREV and LTREV). We report not only coefficient estimates, but also their robust t -statistics within parentheses. The sample period runs from January 15, 2003 to December 29, 2017; with the first subsample ending on June 25, 2010 and the second subsample beginning on June 28, 2010.

	affine conditional alpha			nonparametric conditional alpha		
	entire sample	first half	second half	entire sample	first half	second half
constant	0.0006 (5.50)	0.0006 (3.08)	0.0006 (5.02)	0.0005 (4.40)	0.0005 (2.53)	0.0005 (3.75)
MKT	0.0276 (1.08)	0.0262 (0.69)	0.0438 (2.09)	0.0568 (3.53)	0.0321 (1.48)	0.0704 (3.34)
SMB	-0.1096 (-2.62)	-0.0743 (-1.16)	-0.1867 (-5.28)	-0.0659 (-1.97)	0.0582 (1.20)	-0.2895 (-7.89)
HML	-0.0380 (-0.86)	0.0601 (1.03)	-0.2660 (-5.53)	0.1438 (3.14)	0.2882 (5.17)	-0.2426 (-4.28)
RMW	-0.0089 (-0.17)	-0.0287 (-0.36)	0.0609 (1.04)	0.0274 (0.48)	0.0029 (0.04)	0.0762 (1.07)
CMA	-0.1155 (-1.57)	0.0206 (0.17)	-0.0867 (-1.29)	-0.3368 (-4.85)	-0.1740 (-1.63)	-0.2918 (-3.57)
MOM	0.2722 (11.14)	0.2983 (8.68)	0.2156 (7.78)	0.3023 (11.47)	0.3295 (9.15)	0.2561 (7.52)
STREV	-0.1884 (-6.03)	-0.1895 (-4.69)	-0.1422 (-4.17)	-0.1886 (-5.55)	-0.1815 (-4.24)	-0.1817 (-4.37)
LVREV	-0.0591 (-1.08)	-0.0174 (-0.21)	-0.1330 (-2.14)	-0.3588 (-6.84)	-0.4423 (-5.93)	-0.1755 (-2.67)
R^2	0.18	0.17	0.28	0.20	0.20	0.32
abnormal return	16.00	14.52	16.03	13.03	11.64	13.15